

Protocol-Constrained Residual Estimation for Rigid-Body Inference: Strong Baselines and Visual Controls

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ABSTRACT We present a protocol-constrained residual estimator for short-video rigid-body inference that assigns the main correction to numerical motion and setup features, bounds each update relative to the anchor scale, and evaluates additional visual information using deduplicated numerical inputs and classical tracking controls under the same eight-frame observation budget. On 72 previously held-out synthetic geometry families comprising 1,080 episodes, the locked full estimator reduces held-out-action trajectory error from 1.82315 to 1.45312 mm relative to profiled-dispersion maximum a posteriori estimation, a 20.30% reduction with a post hoc paired 95% interval of $[-0.562, -0.180]$ mm; the improvement is concentrated in sliding, and a separate public pushing evaluation retains 4.38% material-pair improvement when both friction-condition axes are held out within shared object and scene assets. Matched 87-dimensional numerical controls yield 1.44213 mm without an additional view, 1.42632 mm with the frozen Qwen-derived view, and 1.41695 mm with Lucas–Kanade summaries, but neither the Qwen increment nor its difference from tracking is resolved by the two-sided intervals, while the original visual sign and mean tests disagree and real-video transfer remains unsupported after feature-unit normalization.

INDEX TERMS physical parameter estimation, residual learning, rigid-body dynamics, robust estimation, uncertainty evaluation, frozen visual features.

I. INTRODUCTION

SHORT videos can support several explanations of the same motion. Mass, friction, restitution, initial conditions, image tracking errors, and camera geometry interact in an inverse problem. Predicting a plausible property from appearance is therefore different from identifying a property that explains an observed intervention. This distinction matters when an inferred state is used to predict a subsequent action.

Physical prediction and hybrid estimation have been studied through controlled video benchmarks and structured model corrections [1]–[4]. Recent quantitative benchmarks extend this agenda to numerical properties [5], [6]. The unresolved question addressed here is narrower: after a strong numerical path has consumed a short motion observation, does a compact frozen visual feature view provide additional predictive information?

We combine a fixed robust anchor with protocol-specific numerical ridge corrections and a smaller optional vi-

sual residual. The original visual path uses Qwen3-VL-2B-Instruct’s frozen vision component and a previously trained, subsequently frozen observation decoder [7]. It does not generate text. Protocol assumptions restrict which transformed physical coordinates can change, and the generalized-bounce branch retains the anchor. Uncertainty is evaluated separately from the mean correction: the original inflation heuristic is audited, and an additional calibration study uses new RGB geometry families.

The principal contribution is an evaluated residual-estimation design for this restricted inverse problem. The locked estimator improves on profiled-dispersion MAP by 20.30%, with numerical sliding correction explaining most of the gain. The visual question is tested on a strengthened, deduplicated numerical view with classical tracking and pixel controls. This design does not introduce a new ridge solver or establish a general identifiability theorem; its value is the measured accuracy of a constrained correction and explicit accounting for what additional representations contribute.

The study provides three connected results: a strong-anchor comparison with nonlinear optimization and a VLM-free population-prior baseline; matched numerical/visual controls with development-only penalty selection; and an uncertainty/transfer audit with independent calibration families and a corrected real-video feature-unit contract. The separate public pushing experiment tests motion-summary adaptation across friction conditions, rather than material recognition from appearance. Its original 3.90% development result remains below the 10% criterion, and the real-video score remains a transfer failure. All additional analyses of previously evaluated cohorts are exploratory.

II. RELATED WORK

A. QUANTITATIVE EVIDENCE AND IDENTIFIABILITY

Physion and Physion++ evaluate physical prediction, including cases in which latent object properties affect the outcome [1], [2]. Differentiable physical reasoning has also combined video, language, and a physics model [8]. QuantiPhy evaluates numerical physical quantities, while the dynamic-property study of Zhan *et al.* compares classical visual evidence, video-model readouts, and multimodal prompting [5], [6]. These tasks motivate evaluating additional visual information against a strong motion estimator.

Recent preprints examine the difference between supplied evidence and learned priors: Ill-Posed by Design probes evidence use, and EquiSD uses scale equivariance to train metric grounding [9], [10]. Our coordinate mask is a restricted protocol rule. It does not solve general monocular identifiability or establish that all masked coordinates coincide with a measured Jacobian null space.

B. VLM-ASSISTED PHYSICAL INFERENCE

NeRF2Physics connects language-embedded spatial features to physical properties, while PhysX-Anything generates simulation-ready assets from images [11], [12]. These appearance-oriented settings differ from predicting the consequence of a short dynamic intervention.

Phys2Real fuses VLM physical priors with online interaction estimates and evaluates real manipulation [13]. RigPI combines VLM initialization and search-space constraints with force–torque and motion measurements in differentiable simulation [14]. Dynamics uses a structured language representation for video-based rigid-body dynamics [15]. Table 1 identifies methodological distinctions; it is not a performance ranking across incompatible tasks. The Code as Worlds preprint further explores executable representations [16]. Our measured comparisons concern constrained residual estimation; none establishes superiority over these external systems on their original tasks.

C. RESIDUAL ESTIMATION AND FROZEN FEATURES

APHYNITY separates a physical model from a learned correction, and KalmanNet combines structured estimation with neural components [3], [4]. Residual learning and hybrid estimation are thus established ideas. Huber weighting and ridge

regression supply the robust and regularized components used here [17], [18]; Theseus illustrates a broader differentiable-optimization approach [19].

BLIP-2 demonstrates lightweight adaptation of frozen components [20]. V-JEPA 2 and V-JEPA 2.1 provide motion-oriented representations [21], [22]. An additional V-JEPA 2 control uses the same eight observations, numerical anchor, and residual-stage dimension and rules. Its upstream supervision and spatial processing differ from the Qwen-derived decoder view. It is therefore a frozen-representation comparison, not a fully matched intervention on language pretraining or semantic physical knowledge.

D. SELECTIVE USE AND UNCERTAINTY

Answer-Level Trust Selection studies selective use of quantitative physical answers [23]. Our public gate instead scales a vector correction. Neither an average gate value nor lower mean error proves calibrated confidence. Negative log-likelihood (NLL) and the continuous ranked probability score (CRPS) evaluate predictive distributions jointly with coverage and width [24]. Formal conformal guarantees require a compatible procedure and assumptions [25]; the public development-residual radius is empirical, whereas the separate synthetic family-calibration experiment states its sampling target explicitly. The August–September 2026 papers above are explicitly treated as recent preprints, not settled evidence of superiority.

III. PROBLEM AND EVALUATION DEFINITION

The main physical state is

$$\theta = [\log(m/\text{kg}), \log \mu, \logit e]^\top, \quad (1)$$

where m , μ , and e denote mass, dynamic friction, and restitution. An observation model has the form $o = h_p(\theta; u, \eta) + \epsilon$, with known protocol inputs u and nuisance variables η . For ideal horizontal sliding during a fixed contact regime, $a = F/m - \mu g$ explains why distinct known forces can separate mass and friction, whereas unforced deceleration alone does not identify mass. This idealized argument motivates the coordinate permissions in Table 3; it is not a conditioning guarantee for noisy image data.

The two bounce-setup assumptions differ in their treatment of bounce setup information. Generalized translation permits initial height, velocity, and floor nuisance variables with fixed gravity. Setup translation uses the known release height and zero initial velocity and profiles a common vertical offset. Slide inference is shared. Averaging both bounce-setup assumptions is a specified benchmark aggregation, not two independent experimental replications. Camera geometry, motion noise, force, and prior inputs belong to this controlled setup; arbitrary RGB-only deployment is not evaluated.

Protocol permission and final fallback are distinct operations: the bounce protocol permits a restitution coordinate, whereas the generalized-bounce branch subsequently suppresses its entire learned correction. A supplementary audit summarizes the saved initialization Jacobians after nuisance

TABLE 1. Position Relative to Closely Related Methods; Different Sensing and Tasks Preclude a Direct Accuracy Ranking

Method	Information and representation	Role of physical computation	Evaluation boundary
Phys2Real [13]	Visual physical priors and robot interaction estimates	Online adaptation with uncertainty-aware fusion	Real manipulation; different sensing and policy objective
RigPI [14]	VLM semantic initialization, force–torque, and motion	Differentiable parameter refinement in a constrained search space	Real parameter-identification tasks; author metadata reports IROS 2027 acceptance
Δ ynamics [15]	Video and structured textual dynamics representation	Structured dynamics inference	Video-based rigid-body prediction; no matched experiment here
This work	Eight frames, known setup inputs, numerical features, frozen observation-decoder features	Fixed robust anchor and bounded residual heads; no added solver loop	Synthetic analytic actions and an offline public displacement task; real transfer unresolved

TABLE 2. Terminology Used in This Study

Term	Operational meaning
Protocol	Specified motion/contact regime and permitted coordinates.
Bounce-setup assumption	Assumed bounce setup information; slides are shared.
Full model	Numerical and visual residuals, including fixed fallback.
Numerical view	Motion, setup, prior and initialization features.
Analytic action error	Parameter-to-trajectory projection under fixed new inputs.
Harm-guard cell	A setup/protocol point-error harm check.

TABLE 3. Main Protocol Permissions and Assumptions

Protocol	Known inputs and nuisance	Permitted correction
Multiple-force slide	Two force values and switch time; initial position/velocity nuisance	$\log m, \log \mu$
Unforced slide	Zero applied force; initial position/velocity nuisance	$\log \mu$
General bounce	Gravity 9.81 m/s ² ; initial height, velocity, floor nuisance	$\logit e$ permitted by protocol, but final residual disabled
Setup bounce	Known release height 0.65 m and zero initial velocity; free common offset	$\logit e$

projection; it does not equate protocol permissions with the null space of an unrestricted model.

The main action metric uses an independent NumPy implementation of canonical analytic trajectories, not a new PyBullet or PhysX rollout. Estimated and true properties generate trajectories under the same new action, at eight times from 0.125 to 1.0 s. Multiple-force sliding uses forces (0.45, 1.1) N with a 0.35 s switch and initial velocity 0.25

m/s; unforced sliding starts at 1.0 m/s; bounce starts at 0.85 m. Mean absolute position error is averaged within each family/protocol, then equally across three protocols, two bounce-setup assumptions, and 72 families. The images come from synthetic geometry scenes; analytic action utility must be distinguished from engine-rollout accuracy.

The synthetic observation renderer uses PyBullet [26]. The public task predicts a planar displacement vector using an offline onset-aligned video selection. It is a different output from the transformed-property distribution. Its target, timestamp audit, and information-access limitations are defined in Section V.

IV. METHOD

A. FIXED ROBUST ANCHOR AND FEATURE CONSTRUCTION

A fixed Huber-weighted numerical fit produces mean a_{ir} and covariance C_{ir} for episode i and bounce-setup assumption r . Standardized innovations from an initial unit-weight update determine $w_t = \min(1, 1.345/|\nu_t|)$. Observation standard deviations are replaced by $\sigma_t/\sqrt{w_t}$ before the nonlinear fit. These weights are computed once, rather than repeatedly recomputed as a converged Huber M-estimator. The Cauchy control uses $w_t = [1 + (\nu_t/2.385)^2]^{-1}$. The underlying solver retains the specified priors, nuisance handling, and fixed dispersion scale. The covariance is a local curvature approximation; contact/start-selection uncertainty is not integrated out. Detailed code locations are supplied in the supplement.

The numerical view $x_i^N \in \mathbb{R}^{192}$ flattens 24 channels at eight frames: active-axis position, relative position, finite-difference velocity and velocity difference, log observation noise, time, time increment, time from force switch, switch indicator, two forces, switch time, three active-coordinate flags, three prior means, three log prior scales, and three normalized linearization coordinates. It excludes the selected Qwen-derived residual channels, but includes known setup and prior information. It is not an uncontrolled image-only baseline.

All compared primary paths share a frozen appearance prior. A first-image Qwen forward pass with a fixed appearance-description prompt supplies a final-token representation to a learned linear mean/scale head. Thus “numer-

ical only” denotes the residual head’s feature restriction, not an absence of VLM computation throughout the pipeline. The overall gain measures residual adaptation conditional on this common prior; the visual increment isolates the additional eight-frame feature view. The prior stage uses the language-model representation, while the additional visual residual path does not generate text. The supplement distinguishes these upstream stages and their training scope.

The visual path uses the last pre-merge states of the Qwen vision encoder: a 32×32 grid with 1,024 features per frame. Frames are processed separately at a 512^2 pixel budget with BF16 vision computation. A previously trained, frozen observation decoder fuses those states with RGB, tracking, an initial bounding box, and temporal transport. Its attention-weighted 32-channel states, captured before the static/dynamic position heads, supply eight pooled vectors. No text prompt or language-decoder state is used in this extraction. The 48-dimensional residual view concatenates their temporal mean with eight dynamic and eight static position corrections normalized by tracking noise. Thus the features are VLM-derived observation-decoder features, rather than 32 directly selected language-model channels.

B. SEQUENTIAL RESIDUAL TRAINING AND INFERENCE

Let $s_{ir} = \sqrt{\text{diag}(C_{ir})}$. Separate ridge heads are fitted for each protocol and bounce-setup assumption, using only that protocol’s active output coordinates. For a training feature matrix X , each channel is centered and divided by its training standard deviation, floored at 10^{-6} . Let $\tilde{X}$ denote the standardized matrix and $\bar{y}$ the training target mean. Each head solves

$$W_\lambda = \arg \min_W \{ \|\tilde{X}W - (Y - \mathbf{1}\bar{y}^\top)\|_F^2 + \lambda \|W\|_F^2 \}. \quad (2)$$

Its prediction is $f(x) = W_\lambda^\top \tilde{x} + \bar{y}$; the intercept is unpenalized. The loss is a sum, not a sample-averaged loss, which fixes the meaning of the reported penalties. Numerical targets are $(\theta_i - a_{ir}) \oslash s_{ir}$, where $\oslash$ denotes elementwise division. The numerical prediction is clipped before forming the visual target:

$$z_{irp}^N = \text{clip}(f_{rp}^N(x_i^N), -2, 2), \quad (3)$$

$$z_{irp}^V = \text{clip}(f_{rp}^V(x_i^V), -1, 1). \quad (4)$$

Ridge penalties are 1 and 10, respectively. If $a_{ir}^N = a_{ir} + s_{ir} \odot (M_{ip} z_{irp}^N)$, the visual training target is $(\theta_i - a_{ir}^N) \oslash s_{ir}$ on the active coordinates. Thus the visual target uses the numerical head’s *in-sample training prediction*. Feature normalization, target centering, and both fits use the outer training families only. The complete pair is evaluated on the outer held-out family fold. Cross-fitted numerical training targets would define a different estimator and are not used here.

With diagonal protocol mask M_{ip} , the mean update is

$$\hat{\theta}_{ir} = a_{ir} + s_{ir} \odot [M_{ip}(z_{irp}^N + 0.25z_{irp}^V)]. \quad (5)$$

For generalized bounce, development observations of harmful learned correction motivated setting the full correction to

zero; this fallback was fixed before confirmation. Removing the visual term elsewhere retains the numerical correction; it does not return the original anchor. Sequential refers to target construction: both fitted heads can be evaluated from their feature views at inference.

The stored weights and intercepts number $2(2 + 1 + 1)[(192 + 1) + (48 + 1)] = 1,936$. Of these, 242 belong to generalized-bounce heads whose outputs are discarded, leaving 1,694 potentially active coefficients across the other branches. These counts exclude the foundation model, previously trained observation decoder, normalization statistics, and the public extension. They describe residual-stage capacity. On 100 batch-one episodes, median residual-head time is 0.342 ms, whereas the sum of measured inference components is 362.5 ms on an RTX 5080; the supplement specifies the staged timing scope. A development sensitivity sweep retains the original shrinkage 0.25 without confirmation-set reselection. Collapsing repeated setup channels to an 87-dimensional numerical view is reported separately as a post hoc variant.

C. UNCERTAINTY AND INDEPENDENT CALIBRATION

For the accepted mean correction $d = \hat{\theta} - a$, we evaluate

$$\Sigma_\beta = C + \text{diag}[(\beta d)^2]. \quad (6)$$

The original locked setting is $\beta = 0.1$. A development-only CRPS comparison selects $\beta = 0$ for the additional calibration analysis; all original distributional scores are retained as historical results. Positive semidefiniteness alone supplies no calibration or predictive-score justification. NLL and marginal Gaussian CRPS are calculated on each protocol’s active principal covariance submatrix in transformed-property space.

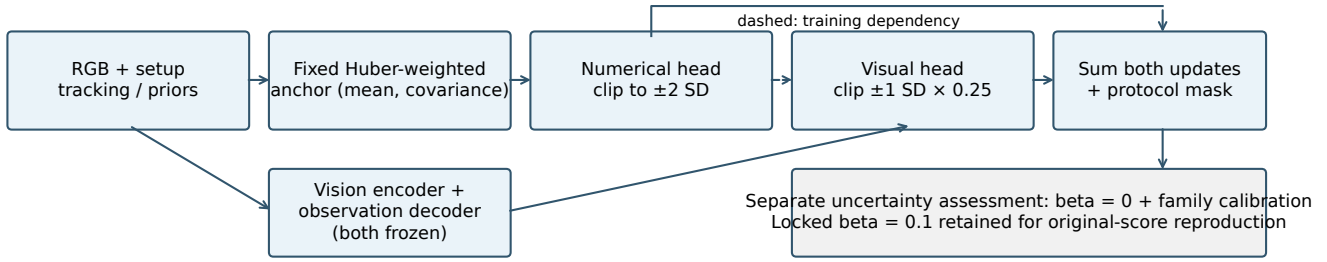
An independent set of 40 newly rendered geometry families (600 RGB episodes) supplies calibration scores after the original full model’s predictions are sealed. One episode and setup assumption per family is selected by a fixed random seed. Its score is the square root of active Mahalanobis error divided by the active-dimensional 90% chi-square threshold. The 37th of 40 scores defines $\hat{q}$, and the calibrated covariance is $\hat{q}^2 C$. Under exchangeable families, this construction targets a randomly sampled episode/setup from a new family; it does not imply simultaneous coverage of every episode. A separately reported maximum-within-family score addresses the latter event. The original 72-family cohort supplies a descriptive evaluation, with no coverage-based reselection.

The clipping constants have an explicit displacement interpretation. Writing $D = \text{diag}(\sqrt{C_{jj}})$ and $d = Dz$ on k permitted coordinates, $|z_j| \leq 2.25$ implies

$$D_{KL}[\mathcal{N}(a + d, C) \|\mathcal{N}(a, C)] \leq \frac{1}{2} k (2.25)^2 \lambda_{\max}(DC^{-1}D). \quad (7)$$

This is an a posteriori bound for the original empirical constants, not a claim that they were selected by a KL optimization. Correlated coordinates prevent replacing the eigenvalue factor by one without further assumptions.

A Main transformed-property estimator



B Separate public displacement task (task-specific fitting)

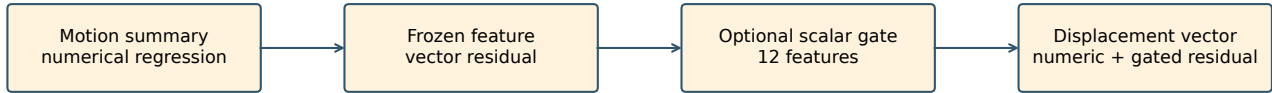


FIGURE 1. Main residual estimator and separate public-task extension. All primary paths share the first-image learned prior supplied to the fixed anchor. Training dependencies are distinguished from prediction outputs. Disabling only visual correction retains numerical adaptation; complete anchor fallback applies to generalized bounce. Calibration is a separate output assessment; the public displacement head has its own empirical radial region.

D. PUBLIC DISPLACEMENT HEAD AND GATE

The public extension predicts a two-dimensional displacement in meters, rather than a physical-property distribution. It fits a ridge-regularized numerical regression with iterative Huber-style residual weighting on an 11-dimensional motion summary, followed by a 144-dimensional visual temporal-feature regression on its in-sample residuals. The selected numerical, visual, and gate penalties are 10, 1,000, and 0.1, respectively. Numerical weighting is recomputed for at most 50 iterations; this differs from the main anchor's fixed innovation weights. Visual corrections are norm-clipped to the development target-norm standard deviation, 0.067724 m for the final fit. A scalar quality gate produces

$$\hat{y}_i = \hat{y}_i^N + g_i \Delta \hat{y}_i^V, \quad 0 \leq g_i \leq 1. \quad (8)$$

Its 12 inputs summarize tracking/decoder disagreement, temporal feature variation, object-size stability, prediction and correction magnitudes, and their alignment. Four-fold cross-fitting inside each of five outer folds generates component predictions used to train an optimal-scalar target:

$$g_i^* = \text{clip} \left(\frac{(y_i - \hat{y}_i^N)^\top \Delta \hat{y}_i^V}{\max(\|\Delta \hat{y}_i^V\|^2, 10^{-12})}, 0, 1 \right). \quad (9)$$

The target minimizes squared vector error along a fixed correction; model selection uses mean Euclidean displacement error, denoted vector mean absolute error (MAE). For episode errors $e_i = \|\hat{y}_i - y_i\|_2$, the reported MAE is $n^{-1} \sum_i e_i$ and root mean squared error (RMSE) is $\sqrt{n^{-1} \sum_i e_i^2}$. Neither is the componentwise mean absolute error. Five gate penalties are compared using the outer predictions. Because component choices and gate penalties were selected using the same development cohort, this is cross-fitting of gate targets, not a fully nested unbiased estimate of all model selection.

The selected out-of-fold radial errors supply the 90% radius, using rank $\lceil (53 + 1)0.9 \rceil = 49$. The same 53 episodes participate in fitting, selection, and residual calibration. The procedure is neither independent split conformal nor an implemented CV+/jackknife+ construction. We therefore report an empirical development-calibrated radial region. The final models and radius are fixed before public confirmation scoring. A valid future split-conformal design would require a calibration partition excluded from fitting and selection. A 20-episode calibration sample would have rank resolution $1/21$ and 90% rank 19; small size makes estimation coarse but does not itself invalidate split conformal. We do not repartition the already evaluated cohort to claim new independent calibration.

V. EXPERIMENTAL PROTOCOL

A. DEVELOPMENT AND NEW-GEOMETRY CONFIRMATION

The residual candidate was selected using 132 development geometry families and six-fold family cross-fitting. Its development action-error reduction was 31.61% against the strongest fixed robust estimator; the visual increment was 1.44%.¹ The local plan required at least 10% full-model improvement, positive paired visual direction, proper-score guards, joint 90% coverage within an author-defined 85–98% range, and bounded width and protocol harm. Candidate configuration and checkpoint were locally locked before generation of the new confirmation cohort. Hashes establish content identity; they are not public preregistration or independent proof of chronology.

The new cohort contains 72 geometry families, 1,080 episodes, and 8,640 frames. All episodes are scored. Stored

¹The development cohort includes the 720 episodes used to train the upstream observation decoder. These six folds isolate residual fitting only and are not equivalent to the new-geometry confirmation.

records report prediction sealing before target opening and no subsequent fitting or exclusions. We replay the saved statistical inputs here; we do not claim a new end-to-end training run or independent reconstruction of every access event.

Controls are unit, Huber, and Cauchy weighting, a frozen legacy observation adapter, an RGB temporal control, and the matched numerical residual. The latter removes the visual head while retaining the anchor, numerical head, protocol mask, and accepted-update covariance rule. Family losses are resampled in 20,000 paired draws. In each draw, the robust reference is the minimum sampled mean among the three fixed robust methods. The original-study rule combines a point improvement of at least 10% with a negative one-sided 95% upper bound; it does not establish a population improvement lower bound of 10%. Six cell guards use the better cell error of the globally strongest robust method and the legacy adapter, with both 10% relative and 0.10 mm absolute harm limits. These are historical engineering criteria, not simultaneous confidence guarantees. Follow-up analyses use two-sided family intervals and paired sign-flip mean tests. Three focal contrasts are reported with Holm adjustment, alongside a global family of 59 available action-error comparisons. Neither adjustment removes the exploratory status of reusing previously evaluated cohorts.

B. OFFLINE PUBLIC DATASET TASK

PhysProbe Push was obtained at the pinned dataset commit identified in the supplement, associated with Isaac Sim 4.5 and Isaac Lab 2.2.1 [27], [28]. A fixed episode selection excluded feasibility episodes 0–9 and assigned 64 candidate episodes to development and 128 to confirmation. An RGB eligibility procedure retained 53 and 123. The split is by episode; shared physical-parameter, object, or environment groups have not been established as independent across these roles.

The saved video headers and parquet timestamps agree on 50 fps for all 176 retained episodes. Frame offsets are $[-21, -16, -11, -6, -1, 4, 9, 14]$ around detected onset. The target is the planar position at onset+90 minus the position at onset. Thus the last observation is 0.28 s after onset, the target endpoint is 1.80 s after onset, and the remaining horizon is 1.52 s. This is displacement from onset, not displacement starting at the last observation. Source frames are resized to 280×280 for the decoder path.

The extraction program tracks the red component across the complete video before determining eligibility and onset. Failure on a later frame can exclude an episode. Although selected features use only eight frames and do not read simulator targets, cohort selection uses future video availability. We therefore describe an offline, onset-aligned, visibility-selected benchmark, not validated causal online forecasting. The timing audit does not eliminate selection bias or establish object-group independence.

The public development search produced a 3.90% improvement and did not pass its original 10% threshold. A

TABLE 4. Held-Out-Action Trajectory Error on 72 Families

Method	MAE (mm)	Versus MAP
Profiled-dispersion MAP	1.82315	reference
Iterated Huber	1.91482	+5.03%
Iterated Cauchy	1.91333	+4.95%
Nuisance-marginal Laplace	1.91803	+5.20%
VLM-free population-prior MAP	1.86647	+2.38%
Locked numerical residual	1.46877	−19.44%
Locked full estimator	1.45312	−20.30%

The original one-step fixed-weight Huber anchor gives 2.07427 mm (29.95% reduction by the full estimator). Unit weighting gives 2.119 mm; legacy and RGB-temporal controls give 2.151 and 2.384 mm. Direct regressors are reported in the supplement. Added anchor comparisons are post hoc.

narrower secondary directional hypothesis and an empirical coverage guard were recorded before the 123 confirmation trajectories were scored. We preserve that history and do not use 4.27% to retroactively pass the development threshold. Episode-bootstrap intervals use 100,000 stored paired draws. Their interpretation assumes episode exchangeability and does not substitute for an unperformed physical-group analysis.

Reporting the failed development criterion makes the subsequent secondary direction interpretable and prevents retrospective threshold adjustment.

VI. RESULTS

A. STRONG ANCHORS AND RESIDUAL ESTIMATION

The locked full estimator obtains 1.45312 mm held-out-action trajectory MAE, versus 1.82315 mm for profiled-dispersion MAP, the lowest-error classical anchor tested here (Table 4). This is a 20.30% reduction; full-minus-MAP difference is -0.37003 mm with a post hoc paired 95% interval $[-0.56199, -0.18024]$ mm and three-contrast Holm-adjusted $p = 0.00150$. These are local numerical fits, not certified global optima. The profiled-MAP convergence rate is 99.86%; unsuccessful fits are retained.

The added Laplace baseline fits a joint Gaussian MAP and marginalizes nuisance in its local Gaussian approximation by a Schur complement. Its marginal mean equals the joint mode; it does not perform non-Gaussian posterior integration. All 1,440 distinct episode/setup fits converge without nuisance-bound contact or rank deficiency. The VLM-free baseline uses the same calibrated RGB tracks, a physical population prior estimated solely from development targets, and observation-derived nuisance starts, without an appearance prior or residual adapter. Its error is 1.86647 mm, with 99.93% convergence. This makes the cost/accuracy comparison explicit while distinguishing a changed prior from removal of only the additional visual head.

The original visual increment is 1.07%, with mean-difference interval $[-0.03114, +0.00053]$ mm including zero. Its exact sign test yields $p = 0.02446$, testing frequency of improvements rather than mean magnitude. These

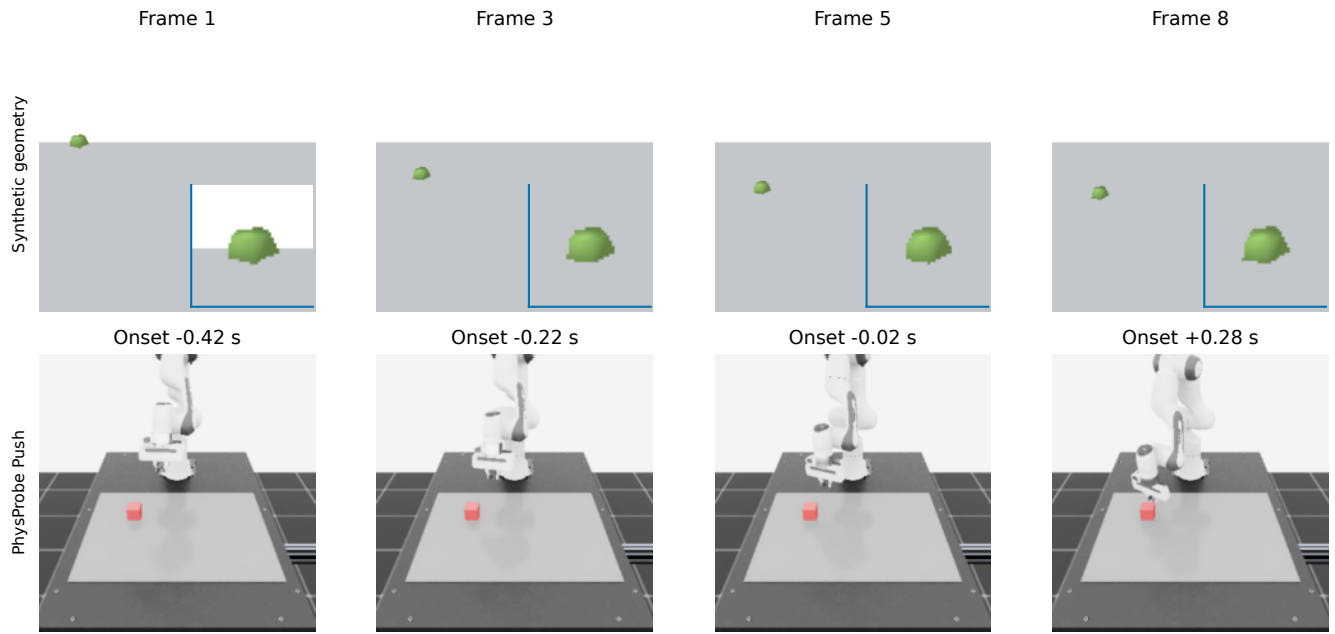


FIGURE 2. Representative inputs from the synthetic geometry and public pushing tasks. Four of eight selected frames are shown; synthetic insets enlarge the tracked object while retaining the complete scene. Public time labels are relative to detected onset at the verified 50 fps cadence; no metric scale is inferred from the image alone.

outcomes disagree as tests of a positive visual result; the sign test does not override the unresolved mean effect. The original local effect-size and harm-guard rules are preserved in the supplement.

a: Where the improvement occurs

Numerical adaptation explains most of the improvement. Table 5 shows that generalized bounce is identical across residual arms because of the fixed fallback, while the substantive changes occur in sliding. The supplement retains nonlinear-solver diagnostics and direct-regression controls.

TABLE 5. Protocol Breakdown in Millimeters; Iterated Huber Is Post Hoc

Protocol	Fixed H.	Iterated H.	Numeric	Full
Multiple forces	3.633	3.068	3.020	2.990
Unforced slide	2.156	2.087	0.978	0.960
General bounce	0.515	0.834	0.515	0.515
Setup bounce	0.352	0.345	0.301	0.304

Against the original cell-specific best-of-robust/legacy reference, sliding contributes 99.9% of the macro error reduction; unforced sliding, whose only corrected coordinate is $\log \mu$, contributes 61.8%. That reference differs from the profiled-MAP reference in Table 4. The six-cell mean and coordinate-specific errors are reproduced in the supplement. The concentration in sliding is compatible with correction of systematic anchor error; the present design does not fully separate this mechanism from acquisition of additional information.

TABLE 6. Deduplicated Numerical and Representation Controls (mm)

Additional view	Numeric dim.	Develop.	Evaluated
None (locked)	192	1.37193	1.46877
Qwen-derived (locked)	192	1.35223	1.45312
None	87	1.36567	1.44213
Qwen-derived	87	1.34792	1.42632
Lucas-Kanade summary	87	1.35202	1.41695
Raw-pixel PCA	87	1.36595	1.43405
V-JEPA 2 PCA	87	1.37007	1.44407

b: Deduplicated numerical view and visual controls

The original 192-dimensional numerical view repeats 15 setup channels at each of eight frames. Keeping those channels once yields 87 dimensions. Its numerical penalty is selected from $\{0.1, 1, 10\}$ on six development family folds; the selected value is 0.1. All added views use the same 87-dimensional numerical fit, 48 additional dimensions, eight frames, masks, clipping and shrinkage, with visual penalties selected only from $\{1, 10, 100\}$ on those development folds. PCA for pixels and V-JEPA is fitted inside each training fold. Development scores are selection scores, not independent generalization estimates.

The 87-dimensional numerical-only cell gives 1.44213 mm, providing the strengthened numerical comparator. Its Qwen-derived increment is -0.01582 mm (1.10%), with interval $[-0.03114, +0.00057]$ mm and focal Holm $p = 0.10139$. Lucas-Kanade summaries give a lower point error; Qwen-minus-tracking difference is $+0.00936$ mm, with interval $[-0.01054, +0.03061]$ mm and focal Holm $p = 0.37728$.

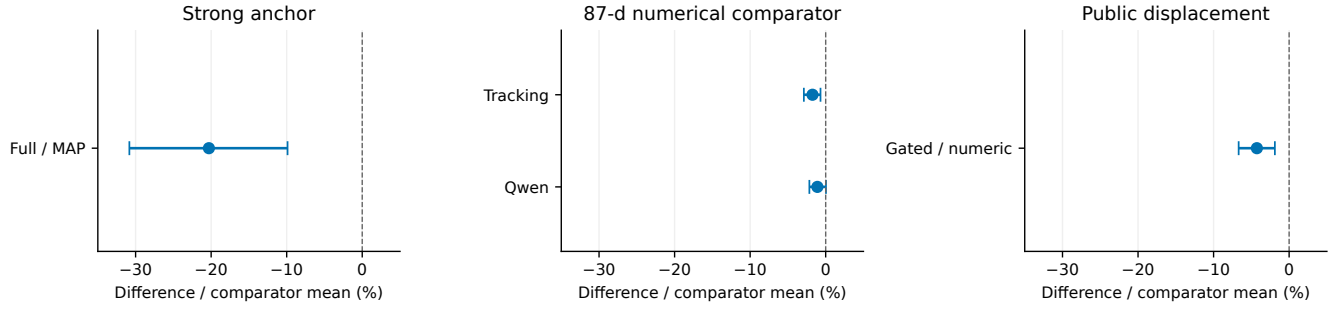


FIGURE 3. Normalized paired effects on one common horizontal scale. Each contrast is divided by its comparator mean error; negative values favor the candidate. Error bars are paired 95% bootstrap intervals. The panels retain distinct estimands and do not pool synthetic trajectory errors with public vector-displacement errors.

These results do not establish Qwen superiority, tracking equivalence, or a universal bound on visual information. The original model is not replaced by the best evaluation arm.

The classical summary tracks initial-cue ROI corners using pyramidal Lucas–Kanade and summarizes displacement, second differences, direction, scale and flow statistics. It retains all episodes, including 1,189 failed transitions among 7,560 evaluation transitions, using a fixed zero-motion fallback and failure feature. It is a concrete non-neural control, not an optimized tracking upper bound. Channel controls in the supplement place most Qwen-derived signal in position corrections. Its decoder has upstream task supervision absent from the PCA controls, so these comparisons do not isolate pretraining effects. The cross-fitted visual-target result (1.45162 mm; interval $[-0.03446, +0.00095]$ mm versus numerical-only) and harmful generalized-bounce correction remain reported.

c: Engine-rollout proxy

A post hoc PyBullet proxy applies the same new actions to estimated and true properties of a nonrotating sphere. Huber, numerical, and full errors are 2.236, 1.620, and 1.606 mm. The full reduction relative to Huber is 28.19%; family-level Spearman correlations with the analytic metric range from 0.961 to 0.986. This supports agreement of rankings under the controlled contact proxy, while the primary metric remains a parameter-to-trajectory projection. It does not validate roll-outs of the original geometry assets or real manipulation.

B. PROPER SCORES AND CONSERVATIVE COVERAGE

Table 7 reports the active transformed-property scores. The full mean-and-covariance estimate improves NLL and CRPS over the robust baseline. Its joint 90% coverage is 97.36%, exceeding nominal coverage by 7.36 percentage points. This is consistent with covariance that is too broad relative to the corrected errors. Passing the 85–98% engineering guard is distinct from matching nominal coverage. The mean joint-axis width ratio is 1.002 relative to Huber. Since the mean and covariance change together, these results do not isolate a causal benefit of the inflation rule.

TABLE 7. Transformed-Property Distribution Evaluation

Method	NLL/dim.	CRPS	90% coverage
Huber	−2.372	0.01755	0.9343
Numerical residual	−2.542	0.01314	0.9736
Numerical + visual	−2.546	0.01311	0.9736

TABLE 8. Public Displacement Error in Millimeters (123 Held-Out Episodes)

Method	MAE	Median	RMSE
Constant prior	132.132	117.007	145.861
Numerical	74.182	63.054	89.584
Raw visual residual	71.210	61.252	84.631
Gated visual residual	71.015	58.338	85.006

a: Independent-family calibration

The development CRPS criterion selects zero inflation. Forty new calibration families give $\hat{q} = 0.4421$; the 37th-order-statistic region covers 81.94% of the original evaluation episodes under equal family/protocol/setup weighting. This does not meet the specified descriptive 88–93% coverage range; no threshold is retuned on evaluation coverage. Protocol-specific coverage and the simultaneous-family alternative are reported in the supplement. Calibration changes uncertainty regions, not point predictions.

C. PUBLIC DISPLACEMENT AND GATE DECOMPOSITION

The public numerical predictor obtains 74.182 mm vector MAE. The raw visual residual yields 71.210 mm, and gating yields 71.015 mm (Table 8). The complete visual-plus-gate increment is 4.27%, with paired difference -3.167 mm and episode-bootstrap 95% interval $[-4.957, -1.400]$ mm. This supports the recorded secondary direction within the selected offline cohort, subject to the episode-exchangeability limitation.

Mean target displacement is 193.869 mm (median 215.989 mm). Numerical, raw, and gated normalized MAEs are 0.383, 0.367, and 0.366; their vector R^2 values are 0.621, 0.662, and 0.659. The development-mean constant prior has $R^2 = -0.0044$, providing a scale sanity check.

The gate alone changes MAE by -0.195 mm (0.274%), with post hoc 95% interval $[-1.785, +1.360]$ mm. Its RMSE increases from 84.631 to 85.006 mm. The gate is therefore an optional quality weighting in these results, not an established source of significant improvement. The raw-visual-versus-numerical interval is $[-5.872, -0.025]$ mm; this additional decomposition is also post hoc and unadjusted for multiple comparisons.

The gate-only change improves 52 episodes, worsens 64, and leaves seven unchanged. Its leave-one-episode-out mean difference ranges from -0.507 to $+0.074$ mm, so even its mean direction is sensitive to an individual episode. By contrast, the complete gated-versus-numerical mean remains negative in every such deletion. These are descriptive post hoc diagnostics; the originally selected gated model is retained without confirmation-set reselection.

Empirical radial 90% coverage is 96.75% for the gated model and 95.93% for numerical prediction. Isotropic Gaussian NLL changes from -2.612 to -2.691 ; the isotropic scale is derived from the fixed radial radius. The mean gate of 0.545 describes correction amplitude, not confidence calibration. External readouts were fitted on PhysProbe development targets, so this is task-specific adaptation to another simulator dataset rather than zero-shot transfer of a physical-property posterior.

D. REAL-VIDEO UNIT CORRECTION

The public real-friction diagnostic contains 33 videos from 11 object-surface families [6]. The original combined score has family Spearman -0.70 , compared with 0.027 for pooled-only and -0.455 for position-only scores. We reproduce those scores exactly, then divide both real pixel-correction channels by an RGB-only tracking-noise estimate, matching the dimensionless form of synthetic corrections. The estimate is a quadratic-detrended median absolute deviation with a one-pixel floor; values span 1.000–1.077 pixels. It is a noise proxy, not measured tracking error.

After correction, family rank correlations remain -0.70 , 0.027 and -0.455 , respectively. The corrected combined interval is $[-0.972, -0.121]$. Thus the documented unit mismatch has been addressed, but this normalization does not restore transfer. Most scales lie at the floor, so the intervention is small; camera projection and decoder-domain differences remain. Neither score is a physical friction estimate. The 152-frame bridge and scalar proxy use deceleration divided by initial speed, which already cancels pixel units: rescaling numerator and denominator reproduces their correlations of 0.355 and 0.427. Their intervals include zero. Full before/after values and the unit contract are provided in the supplement.

E. ADDITIONAL MATERIAL-DISJOINT PUBLIC EVALUATION

A metadata audit identifies 16 object-friction pairs and 16 surface-friction pairs in PhysProbe Push. The original eligible development and evaluation sets share 15 object-material and all 16 surface-material conditions. Thus the original episode

TABLE 9. New-Episode Evaluation with Both Material Axes Held Out (329 Episodes)

Method	Material-pair MAE	Episode MAE
Numerical	63.419	65.091
Raw visual residual	61.943	64.642
Gated visual residual	60.641	62.753

Both columns are in millimeters. Material-pair MAE is the primary metric and weights observed material combinations equally.

Episode MAE is descriptive. Episode-normalized errors are 0.342, 0.340, and 0.330; vector R^2 values are 0.634, 0.649, and 0.657.

Mean target displacement is 190.281 mm.

split is not material-disjoint. All episodes use the same red cube; friction conditions are metadata scalars without corresponding material-identifying appearance cues in this setup. The result tests generalization of motion summaries across friction conditions, not recognition of material from appearance or transfer to distinct object/scene assets.

We fixed a follow-up protocol before reading new videos or displacement targets. It excludes all 202 previously selected or feasibility episodes and assigns eight object materials and eight surface materials to evaluation. Training uses the complementary materials on both axes; mixed assignments are omitted. The unchanged visibility procedure retains 285 training and 329 evaluation episodes from 320 and 348 candidates. The evaluation spans 63 of 64 possible held-out material pairs. No object- or surface-friction pair is shared between fitting and evaluation, and no complete video hash is shared. The original architecture and penalties are retained, with newly fitted coefficients. Internal gate targets also exclude both materials of each held-out pair. Evaluation trajectories are downloaded only after predictions are sealed.

Equal weighting of the 63 material-pair mean errors yields 63.419 mm for numerical prediction and 60.641 mm for the gated residual, a 4.38% reduction (Table 9). The paired difference is -2.778 mm, with a two-way material-bootstrap 95% interval of $[-6.044, -0.116]$ mm. Marginal differences favor the candidate in seven of eight object-material groups and five of eight surface-material groups. These outcomes satisfy the locally fixed directional and group-consistency rule. The interval's upper endpoint is close to zero, and eight groups per axis limit precision. This validates the fitted readout under held-out friction conditions within the same cube/scene setting. Architecture selection previously used episodes from this material bank; the study does not establish historically untouched material discovery, new object-asset transfer, or a causal online evaluation. The original 3.90% development failure and 4.27% episode-split result remain separate records.

F. COST AND INTERPRETATION

The locked additional visual path raises recorded median component totals from 129.1 to 362.5 ms (2.81 times), while its residual heads take 0.342 ms. The shared first-image appearance prior itself takes 68.4 ms. These staged,

TABLE 10. Evidence Established and Limits of the Present Design

Supported within the tested setting	Not established
Sliding residual adaptation improves profiled MAP and Laplace anchors.	Separation of anchor bias correction from additional information.
Deduplicated numerical and classical tracking controls are evaluated.	Qwen superiority over tracking or isolated pretraining benefit.
Refitting generalizes across both friction-condition axes.	Transfer to unseen object or scene assets.
Independent geometry-family calibration is separately assessed.	Nominally calibrated posterior coverage.
The real-video score fails after noise-unit normalization.	A causal explanation or general impossibility of real transfer.

model-resident sums are not contiguous cold-start latency. The matched residual arms retain that prior; the separate population-prior MAP row in Table 4 is VLM-free. Parameter and timing comparisons in the supplement distinguish residual coefficients, upstream decoder supervision, PCA storage and feature-extraction cost. No speed advantage is inferred by comparing concurrent CPU extraction with staged GPU totals.

VII. DISCUSSION

The supported contribution is a constrained residual estimator whose trajectory-projection improvement persists against the tested profiled-MAP and Laplace anchors. The evidence is strongest for sliding corrections, not a broad rigid-body reasoning framework. Duplicate setup features change effective ridge regularization; removing them strengthens the numerical comparator without eliminating the small, unresolved Qwen-derived mean increment. A classical tracking summary reaches a lower point error at the same residual dimension. Additional visual computation therefore requires task-specific justification rather than an assumption that a foundation representation is necessary.

The protocol mask expresses structural assumptions of the simplified inverse problem. Under unforced sliding, acceleration is independent of mass; known distinct forces can separate mass and friction; restitution is exposed only through the specified bounce model and nuisance assumptions. These permissions agree with elementary system-identification reasoning, but the cached initialization Jacobian is not a final-solution Fisher proof. One-column condition numbers are identically one and supply no conditioning evidence. We make no automatic-mask equivalence or new identifiability-theorem claim.

The analytic metric is a parameter-to-trajectory projection, and the controlled PyBullet proxy shares idealized assumptions. Neither provides a model-misspecification stress test on original object assets. PhysProbe retains the same cube/scene and earlier material-bank architecture-selection history; its holdout tests friction-condition generalization of motion summaries. Matched upstream supervision, text-based external-method ports, and multi-asset/physics/sensor shift evaluations remain outside the demonstrated scope. Table 10 records these boundaries. The unit-corrected real failure and any unsuccessful calibration target are retained rather than converted

into evidence of impossibility.

VIII. CONCLUSION

Protocol-constrained residual estimation improves the locked full model's held-out-action trajectory error by 20.30% relative to the strongest tested profiled-dispersion MAP anchor, primarily through numerical sliding correction. Deduplicated controls and a classical tracker identify how much improvement remains after strengthening the numerical view; the tested Qwen-derived representation does not show a resolved advantage over tracking. Independent-family calibration and corrected real-video diagnostics separate distributional quality and transfer from point-estimation accuracy. The resulting claim is a useful estimator within a controlled regime, with explicit evidence needed before extending it to object-asset shifts, mismatched physics or real-world physical-property inference.

DATA AND CODE AVAILABILITY

Verification materials are provided in release v0.4.0 of <https://github.com/Jinhong-Yang/physics-residual-verification>. The Overleaf project includes source, figure data, selected observations and predictions, residual checkpoints, and executable calculation/refitting tools. Strong-anchor, representation, calibration and unit-corrected real-video analyses include plans, numerical inputs, predictions and replay tools. Additional comparisons use previously evaluated cohorts and are explicitly exploratory. The RGB companion reproduces eight complete inference paths, including the shared appearance prior, after separately obtaining hash-checked foundation weights. Recorded latency summaries can be replayed; fresh hardware measurement additionally needs the measurement source's original 100-episode RGB inputs and upstream environment. Full upstream training is not reproduced. The public dataset is available at the pinned repository in [28].

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REFERENCES

- [1] D. M. Bear, E. Wang, D. Mrowca, F. J. Binder, H.-Y. Tung, R. T. Pramod, C. Holdaway, S. Tao, K. Smith, F.-Y. Sun, F.-F. Li, N. Kanwisher, J. B. Tenenbaum, D. L. K. Yamins, and J. E. Fan, "Physion: Evaluating physical prediction from vision in humans and machines," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1, 2021.
- [2] H.-Y. Tung, M. Ding, Z. Chen, D. Bear, C. Gan, J. B. Tenenbaum, D. Yamins, J. Fan, and K. Smith, "Physion++: Evaluating physical scene understanding that requires online inference of different physical properties," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.
- [3] Y. Yin, V. Le Guen, J. Dona, E. de Bézenac, I. Ayed, N. Thome, and P. Gallinari, "Augmenting physical models with deep networks for complex dynamics forecasting," in *International Conference on Learning Representations (ICLR)*, 2021. [Online]. Available: <https://arxiv.org/abs/2010.04456>
- [4] G. Revach, N. Shlezinger, X. Ni, A. López Escoriza, R. J. G. van Sloun, and Y. C. Eldar, "KalmanNet: Neural network aided kalman filtering for partially known dynamics," *IEEE Transactions on Signal Processing*, vol. 70, pp. 1532–1547, 2022.
- [5] P. Li, T. Xiang, E. Mao, S. Wei, X. Chen, A. Masood, F.-F. Li, and E. Adeli, "QUANTIPHY: A quantitative benchmark evaluating physical reasoning abilities of vision-language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2026, pp. 33 174–33 184.
- [6] G. Zhan, X. Ma, W. Xie, and A. Zisserman, "Inferring dynamic physical properties from video foundation models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2026, pp. 5214–5224.
- [7] S. Bai, Y. Cai, R. Chen et al., "Qwen3-VL technical report," *arXiv preprint arXiv:2511.21631*, 2025.
- [8] M. Ding, Z. Chen, T. Du, P. Luo, J. B. Tenenbaum, and C. Gan, "Dynamic visual reasoning by learning differentiable physics models from video and language," in *Advances in Neural Information Processing Systems*, vol. 34, 2021.
- [9] B. Meivar, S. Perek, S. Shvartzman, E. Schwartz, and S. Avidan, "Ill-posed by design: Probing evidence use in VLMs," 2026, preprint, June 2026. [Online]. Available: <https://arxiv.org/abs/2606.24335>
- [10] K. Tan, Y. Feng, H. Du, S. Tao, X. Xu, and H. Hong, "Teaching vision-language models to use the scale they are given: Label-free equivariance training for metric physical reasoning," 2026, preprint, September 2026. [Online]. Available: <https://arxiv.org/abs/2609.00658>
- [11] A. J. Zhai, Y. Shen, E. Y. Chen, G. X. Wang, X. Wang, S. Wang, K. Guan, and S. Wang, "Physical property understanding from language-embedded feature fields," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 28 296–28 305.
- [12] Z. Cao, F. Hong, Z. Chen, L. Pan, and Z. Liu, "PhysX-Anything: Simulation-ready physical 3d assets from single image," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, public manuscript: arXiv:2511.13648. [Online]. Available: <https://arxiv.org/abs/2511.13648>
- [13] M. Wang, S. Tian, A. Swann, O. Shorinwa, J. Wu, and M. Schwager, "Phys2Real: Fusing VLM priors with interactive online adaptation for uncertainty-aware sim-to-real manipulation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026, public manuscript: arXiv:2510.11689v2. [Online]. Available: <https://arxiv.org/abs/2510.11689>
- [14] X. He, R. Zhang, W. Jiang, and W. Xu, "RigPI: Dynamic parameter identification of rigid body via VLM-seeded differentiable simulation," 2026, arXiv:2606.25212v2; author metadata reports acceptance to IROS 2027. [Online]. Available: <https://arxiv.org/abs/2606.25212>
- [15] C.-H. Kao, C. P. Huynh, C.-Y. Wang, N. Vedapant, S. Stojanov, B. Hariharan, O. Obiednikov, and N. Zhou, "Dynamics: Language-based representation for inferring rigid-body dynamics from videos," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, pp. 42 364–42 374. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/html/Kao_Dynamics_Language-Based_Representation_for_Infering_Rigid-Body_Dynamics_From_Videos_CVPR_2026_paper.html
- [16] H. Wang, Y. Cai, W. Chen, J. Chi, H. Sun, Q. Dai, Y.-H. Hung, X. Guo, J. Ren, R. Yao, Z. Liu, M. Long, Y. Duan, J. Gao, J. Lyu, F. Liu, and J. Wu, "Code as worlds: Agentic discovery of executable world representations for physical reasoning," 2026, preprint, August 2026. [Online]. Available: <https://arxiv.org/abs/2608.27549>
- [17] P. J. Huber, "Robust estimation of a location parameter," *Ann. Math. Statist.*, vol. 35, no. 1, pp. 73–101, 1964.
- [18] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
- [19] L. Pineda, T. Fan, M. Monge, S. Venkataraman, P. Sodhi, R. T. Q. Chen, J. Ortiz, D. DeTone, A. Wang, S. Anderson, J. Dong, B. Amos, and M. Mukadam, "Theseus: A library for differentiable nonlinear optimization," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.
- [20] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. 40th Int. Conf. Mach. Learn.*, ser. Proc. Mach. Learn. Res., vol. 202, 2023, pp. 19 730–19 742.
- [21] M. Assran, A. Bardes, D. Fan et al., "V-JEPA 2: Self-supervised video models enable understanding, prediction and planning," 2025. [Online]. Available: <https://arxiv.org/abs/2506.09985>
- [22] L. Mur-Labadia, M. Muckley, A. Bar, M. Assran, K. Sinha, M. Rabbat, Y. LeCun, N. Ballas, and A. Bardes, "V-JEPA 2.1: Unlocking dense features in video self-supervised learning," 2026, version 3, June 2026. [Online]. Available: <https://arxiv.org/abs/2603.14482>
- [23] R. Yu, K. Niu, and F. He, "Answer-level trust selection for physical vision-language reasoning," 2026, preprint, August 2026. [Online]. Available: <https://arxiv.org/abs/2608.19807>
- [24] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *J. Amer. Statist. Assoc.*, vol. 102, no. 477, pp. 359–378, 2007.
- [25] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Found. Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2023.
- [26] E. Coumans and Y. Bai, "PyBullet quickstart guide," *Bullet Physics*, 2022, [Online]. Available: https://github.com/bulletphysics/bullet3/blob/master/docs/pybullet_quickstartguide.pdf.
- [27] M. Mittal, P. Roth, J. Tigue, A. Richard et al., "Isaac Lab: A gpu-accelerated simulation framework for multi-modal robot learning," *arXiv preprint arXiv:2511.04831*, 2025.
- [28] leesangoh, "PhysProbe Dynamics Probing Dataset," *Hugging Face Datasets*, 2026, commit b679fcf0a800d60fd05064304e0e8777392f34dd. [Online]. Available: <https://huggingface.co/datasets/leesangoh/dynamics-probing>.



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